

Investment Subsidy in a Heterogeneous Firm Model

1 Model Solution

1.1 Optimal Labor and Reduced-Form Profit

Each firm j solves a static labor choice each period:

$$h_{j,t}^* = \arg \max_{h \geq 0} \{ z_{j,t} k_{j,t}^\alpha h^\nu - wh \}.$$

The first-order condition $\nu z_{j,t} k_{j,t}^\alpha (h^*)^{\nu-1} = w$ yields the closed-form solution:

$$h_{j,t}^* = \left(\frac{\nu z_{j,t} k_{j,t}^\alpha}{w} \right)^{\frac{1}{1-\nu}}.$$

Substituting h^* back into the profit expression, and defining $\pi(z, k)$ as the reduced-form profit function:

$$\pi(z_{j,t}, k_{j,t}) = (1 - \nu) \left(\frac{\nu}{w} \right)^{\frac{\nu}{1-\nu}} z_{j,t}^{\frac{1}{1-\nu}} k_{j,t}^{\frac{\alpha}{1-\nu}}.$$

With calibrated values $\alpha = 0.30$, $\nu = 0.60$, $w = 1$, this simplifies to $\pi(z, k) = 0.4 \cdot (0.6)^{1.5} \cdot z^{2.5} \cdot k^{0.75}$.

1.2 Bellman Equation

Let $V(k, z)$ denote the value function of a firm with capital k and productivity z . The firm's problem is:

$$V(k, z) = \max_{i, k'} \left\{ \pi(z, k) - p(i; \tau) \cdot i - c(i, k) + \frac{1}{1+r} \mathbb{E}_{z'|z} [V(k', z')] \right\} \quad (1)$$

subject to the law of motion $k' = (1 - \delta)k + i$, where:

$$c(i, k) = \frac{\gamma}{2} \left(\frac{i}{k} \right)^2 k + Fk \cdot \mathbf{1}\{i \neq 0\}, \quad (2)$$

$$p(i; \tau) = (1 - \tau) \cdot \mathbf{1}\{i \geq 0\} + p_s \cdot \mathbf{1}\{i < 0\}. \quad (3)$$

The convex component of $c(i, k)$ penalizes rapid capital adjustment with coefficient $\gamma > 0$. The fixed cost $F > 0$ applies whenever the firm invests or sells ($i \neq 0$), creating a region of inaction. The price of investment $p(i; \tau)$ introduces partial irreversibility: firms buy capital at price $1 - \tau$ (net of subsidy) but can only sell at price $p_s \leq 1$.

1.3 Numerical Method

Grids. The productivity process $\ln z_{t+1} = \rho \ln z_t + (1 - \rho) \ln \tilde{z} + \varepsilon_{t+1}$ is discretized with $N_z = 50$ grid points and $m = 3$ standard deviations. The normalizing constant $\tilde{z} = \exp(-\sigma^2/(2(1 - \rho^2)))$ ensures $\mathbb{E}[z] = 1$. Capital is discretized over a log-spaced grid of $N_k = 300$ points on $[0.1, 20.0]$.

Value Function Iteration. We solve (1) by iterating on the Bellman operator. At each state (k_{ik}, z_{iz}) :

1. Compute the expected continuation value $\mathbb{E}V(k', z') = \sum_{iz'} \Pi_{iz, iz'} V(k_{ik'}, z_{iz'})$ for all k' on the grid.
2. Search over all N_k candidate values of k' to find the optimal $i^* = k' - (1 - \delta)k$.
3. Separately evaluate the inaction option $i = 0$, where $k' = (1 - \delta)k$, using linear interpolation.
4. Take the maximum over the grid search and the inaction option.

The inaction option is handled explicitly because the fixed cost creates a discontinuity: at $i = 0$ the firm avoids paying Fk , but $(1 - \delta)k$ generally falls off the capital grid, requiring interpolation. Iteration continues until $\|V^{n+1} - V^n\|_\infty < 10^{-6}$ or 2000 iterations.

Stationary Distribution. Given policy $i^*(k, z)$, we compute the stationary distribution $\mu(k, z)$ by iterating the distribution forward. Mass is split between adjacent grid points using linear interpolation weights to avoid discretization artifacts. The iteration runs until $\|\mu^{n+1} - \mu^n\|_\infty < 10^{-8}$.

2 Model Exploration

We progressively add friction components to understand their individual effects. All explorations use $\gamma = 0.5$, $N_k = 50$, $N_z = 10$.

2.1 Convex Costs Only ($F = 0$, $p_s = 1$)

Without fixed costs or irreversibility, adjustment is smooth and symmetric. Table 1 shows that the inaction rate is only 10.6%: firms continuously smooth capital toward its frictionless target, so the mass near $i/k = 0$ is modest. Negative investment is frequent (33.7%) because with $p_s = 1$ there is no cost asymmetry—selling capital is equally cheap as buying. The positive spike rate (21.7%) reflects the convex cost penalizing large investments but not precluding them.

2.2 Adding Fixed Costs ($F = 0.02$, $p_s = 1$)

Fixed costs create a region of inaction: it is only optimal to pay Fk when the productivity gap from the target capital is large enough. The inaction rate jumps to 66.6%, closely generating the lumpy investment patterns documented empirically. Because the fixed cost applies symmetrically to positive and negative investment, both positive and negative spike rates increase as firms wait and then adjust in large discrete steps. The negative investment fraction falls sharply from 33.7% to 6.1%, as the fixed cost discourages small disinvestments.

2.3 Full Model ($F = 0.02$, $p_s = 0.9$)

Introducing partial irreversibility ($p_s = 0.9 < 1$) makes selling capital more costly. Firms become even more reluctant to disinvest, further reducing the negative investment fraction to 3.5%. The inaction rate rises modestly to 71.6%, as firms now tolerate a wider capital gap before selling. The asymmetry between positive and negative spikes becomes visible: negative spike rates remain low (3.5%), while positive spikes are more common.

Table 1: Model Exploration: Moments Under Alternative Specifications

Moment	Convex Only ($F = 0$, $p_s = 1$)	+ Fixed Costs ($F = 0.02$, $p_s = 1$)	Full Model ($F = 0.02$, $p_s = 0.9$)
Avg. Investment Rate	0.1050	0.1062	0.1011
Inaction Rate	0.1059	0.6662	0.7157
Neg. Investment Fraction	0.3368	0.0605	0.0349
Pos. Spike Rate (>20%)	0.2166	0.2629	0.2390
Neg. Spike Rate (<-20%)	0.0470	0.0605	0.0349

The exploration confirms that the policy functions have economically sensible shapes. In the convex-only case, $i^*(k, z)$ is smooth and monotonically decreasing in k for given z (high-capital firms disinvest, low-capital firms invest). Adding fixed costs produces a band of inaction around the target capital: the policy function is zero for a range of k values

and then jumps to positive or negative adjustment at the boundaries. Irreversibility shifts the inaction region asymmetrically upward—firms are willing to hold excess capital rather than sell it at a discount.

3 Grid Sensitivity Analysis

Table 2 reports how key moments change as the grid becomes finer, using the default parameters ($\gamma = 0.3$, $F = 0.01$, $p_s = 0.7$).

Table 2: Grid Sensitivity: Average Investment Rate and Inaction Rate

N_k	N_z	Avg. Inv. Rate	Inaction Rate
30	8	0.1030	0.7206
50	12	0.0987	0.7395
80	15	0.0979	0.7455
120	20	0.0974	0.7532
200	30	0.0972	0.7529

The moments stabilize between $N_k = 120$ and $N_k = 200$, with changes of less than 0.1 percentage point. This confirms that $N_k = 300$, $N_z = 50$ used in the final analysis provides adequate accuracy. The slight downward trend in the average investment rate as the grid refines reflects the fact that a coarser capital grid overstates returns to investment by not penalizing overshooting.

4 Calibration

4.1 Externally Calibrated Parameters

The production and productivity parameters are set to standard values: $\alpha = 0.30$, $\nu = 0.60$, $\delta = 0.08$, $r = 0.04$, $w = 1$, $\rho = 0.90$, $\sigma = 0.12$, and $\tilde{z} = \exp(-\sigma^2/(2(1 - \rho^2))) \approx 0.931$ (so that $\mathbb{E}[z] = 1$).

4.2 SMM Estimation of Adjustment Cost Parameters

We estimate $\theta = (\gamma, F, p_s)$ by minimizing the weighted distance between model and data moments:

$$\hat{\theta} = \arg \min_{\theta} [m^{\text{model}}(\theta) - m^{\text{data}}]' W [m^{\text{model}}(\theta) - m^{\text{data}}],$$

where m^{data} is drawn from Cooper and Haltiwanger (2006) (Table 3) and $W = I$ (identity matrix). We use a two-stage procedure:

1. **Grid search:** Evaluate the objective over a $6 \times 5 \times 8 = 240$ -point grid covering $\gamma \in \{0.5, 1, 2, 3, 5, 8\}$, $F \in \{0, 0.001, 0.005, 0.01, 0.02\}$, $p_s \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95\}$, using a coarse grid ($N_k = 60$, $N_z = 12$) for speed.
2. **Nelder-Mead refinement:** Starting from the best grid point, apply the Nelder-Mead simplex algorithm (1000 iterations, tolerance 10^{-5}) to refine the estimate.

4.3 Estimated Parameters and Moment Fit

The estimated parameters are $\hat{\gamma} = 0.999$, $\hat{F} = 0.0003$, $\hat{p}_s = 1.000$. Table 3 compares model moments evaluated at these parameters (using the fine grid $N_k = 300$, $N_z = 50$) to the data.

Table 3: Moment Fit: Data vs. Model at Estimated Parameters

Moment	Data	Model	Deviation
Avg. Investment Rate	0.1220	0.0915	-0.031
Inaction Rate	0.0810	0.1367	+0.056
Neg. Investment Fraction	0.1040	0.2436	+0.140
Pos. Spike Rate	0.1800	0.2156	+0.036
Neg. Spike Rate	0.0140	0.0087	-0.005

Discussion of fit. The model broadly replicates the qualitative patterns but faces quantitative limitations. The negative investment fraction (24.4% model vs. 10.4% data) is the largest discrepancy. This reflects a fundamental tension: matching the low Neg. Investment Fraction requires a low resale price p_s (strong irreversibility), but the optimizer drives $\hat{p}_s \rightarrow 1$ because doing so reduces the overall objective. The reason is that $p_s < 1$ worsens the fit on the Inaction Rate and Neg. Spike Rate, which collectively dominate the identity-weighted objective. In other words, the five data moments impose conflicting requirements on p_s , and the model cannot simultaneously satisfy all of them.

5 Policy Functions

Table 4 presents $i^*(k, z)/k$ at selected (k, z) pairs under the baseline calibration. The policy function exhibits the following properties consistent with economic theory:

Table 4: Investment Rate Policy $i^*(k, z)/k$ at Calibrated Parameters ($\hat{\gamma} = 0.999$, $\hat{F} = 0.0003$, $\hat{p}_s = 1.0$)

	$z = 0.42$	$z = 0.63$	$z = 0.98$	$z = 1.42$	$z = 2.20$
$k = 0.10$	0.0000	0.0000	1.0759	2.4167	4.9627
$k = 0.37$	-0.2547	0.0000	0.5053	1.3395	2.9936
$k = 1.40$	-0.2664	-0.1397	0.1726	0.7224	1.7301
$k = 5.29$	-0.3111	-0.2307	-0.0209	0.3169	0.9393
$k = 20.00$	-0.3323	-0.2779	-0.1534	0.0624	0.0800

- **Decreasing in k for fixed z :** High-capital firms face lower marginal returns and tend to disinvest (depreciate capital). Low-capital firms face high marginal returns and invest aggressively.
- **Increasing in z for fixed k :** Higher productivity raises the marginal value of capital, inducing more investment.
- **Inaction at $(k = 0.10, z = 0.42)$ and $(k = 0.10, z = 0.63)$:** Small, low-productivity firms find it optimal not to invest—a region of inaction generated by the small fixed cost.
- **Large positive spikes for small k , high z :** Investment rates above 100% are common when highly productive firms have very little capital; the convex cost prevents full instantaneous adjustment but large investment is still optimal.

6 Stationary Distribution

The stationary distribution $\mu(k, z)$ is concentrated in the interior of the capital-productivity state space. Key features:

- The distribution is right-skewed over k : most firms have moderate capital ($k \in [1, 6]$), with a thin tail of large firms.
- There is positive covariance between k and z : more productive firms tend to hold more capital, yielding $\text{Corr}(k, z) = 0.690$ at baseline.
- The investment rate distribution is right-skewed and unimodal.

7 Policy Analysis: Investment Subsidy

We compare two economies: the baseline ($\tau = 0$) and a subsidized economy ($\tau = 0.10$), using the calibrated parameters. The subsidy reduces the purchase price of capital to $1 - \tau = 0.90$, leaving the resale price p_s unchanged.

7.1 Investment Moments

Table 5: Investment Moments: Baseline vs. Subsidy

Moment	$\tau = 0$	$\tau = 0.10$	% Change
Avg. Investment Rate	0.0915	0.0935	+2.1%
Inaction Rate	0.1367	0.0044	−96.8%
Neg. Investment Fraction	0.2436	0.3046	+25.1%
Pos. Spike Rate	0.2156	0.2324	+7.8%
Neg. Spike Rate	0.0087	0.0116	+33.4%

The most striking effect is the near-complete elimination of inaction (from 13.7% to 0.4%). By lowering the cost of purchasing capital, the subsidy shifts the inaction threshold: it is now optimal to invest in states where firms previously remained inactive. The average investment rate rises modestly (+2.1%) because the extensive margin (number of investors) increases substantially, but many newly-investing firms choose small investment rates. The positive spike rate rises (+7.8%) and the negative investment fraction also increases (+25.1%)—firms that now invest more aggressively in response to the subsidy must eventually disinvest when productivity falls back.

7.2 Aggregate Effects

Table 6: Aggregate Effects of the Investment Subsidy

Aggregate	Baseline ($\tau = 0$)	Subsidy ($\tau = 0.10$)	% Change
Capital $K = \int k d\mu$	4.233	5.407	+27.7%
Labor $H = \int h^* d\mu$	1.238	1.471	+18.9%
Output $Y = \int y d\mu$	2.063	2.452	+18.9%

The subsidy generates large aggregate effects. Capital rises by 27.7%, reflecting the shift in the stationary distribution toward higher capital levels. Labor and output both rise by 18.9%, driven by the complementarity between capital and labor in the Cobb-Douglas production function ($h^* \propto k^{\alpha/(1-\nu)}$). These are long-run, steady-state effects, not transition dynamics.

7.3 Cross-Sectional Patterns

The correlation between capital and productivity increases from 0.690 to 0.719 under the subsidy. The subsidy lowers the effective cost of capital, making it more worthwhile

for high-productivity firms to expand aggressively. Since the convex adjustment cost means that low-productivity firms also invest more (they were previously inactive), the distribution spreads out, but the high- z /high- k correlation strengthens on net. This suggests the subsidy improves allocative efficiency: capital is more concentrated among productive firms.

7.4 Subsidy Cost

The total cost of the subsidy is:

$$\text{Cost} = \tau \int i^*(k, z) \cdot \mathbf{1}\{i^* > 0\} d\mu = 0.0241 \cdot Y_{\tau=0.10},$$

or 2.41% of output. The output gain is 18.9%, yielding a rough benefit-to-cost ratio of about $18.9/2.41 \approx 7.8$. The subsidy appears broadly cost-effective in the partial equilibrium model.

7.5 Interaction with Adjustment Costs

Convex costs. The quadratic adjustment cost limits how quickly firms respond to the lower capital price. Even without a fixed cost, the subsidy would raise investment rates smoothly. With the convex cost, the main effect is an increase in the fraction of firms that invest at all (extensive margin), rather than a large increase in individual investment rates (intensive margin).

Fixed costs. The fixed cost $\hat{F} \approx 0$ in our calibration means that the inaction region is small at baseline (13.7%) and nearly eliminated by the subsidy (0.4%). With a larger F , the subsidy's impact on the inaction rate would be even more dramatic—it directly offsets the fixed cost of initiating investment, so more states cross the threshold from inaction to action.

Irreversibility. With $\hat{p}_s = 1$ in our calibration, irreversibility plays no direct role. In a model with $p_s < 1$, the subsidy would interact with irreversibility by making positive investment relatively more attractive compared to disinvestment, further tilting the distribution toward positive investment. The increase in negative investment and negative spikes we observe (+25.1% and +33.4%) paradoxically arises because the higher capital stock in the subsidized steady state must sometimes be reduced when productivity falls—more capital in the system means more potential for downward adjustment.

8 Accuracy and Convergence

VFI convergence. Table 7 reports convergence diagnostics for the main model configurations. All runs converge within 330 iterations to the tolerance 10^{-6} .

Table 7: VFI Convergence ($N_k = 300$, $N_z = 50$, tolerance 10^{-6})

Specification	Iterations	Final $\ V^{n+1} - V^n\ _\infty$
Convex only ($F = 0$, $p_s = 1$)	330	9.75×10^{-7}
Fixed costs ($F = 0.02$, $p_s = 1$)	326	9.83×10^{-7}
Full model ($F = 0.02$, $p_s = 0.9$)	326	9.99×10^{-7}
Baseline calibrated ($\tau = 0$)	322	9.66×10^{-7}
Subsidized ($\tau = 0.10$)	327	9.90×10^{-7}

Grid sensitivity. As shown in Table 2, the average investment rate changes by less than 0.1 percentage point between $N_k = 120$ and $N_k = 200$, and the inaction rate stabilizes similarly. The fine grid $N_k = 300$, $N_z = 50$ used for final results is well beyond the convergence threshold.

Limitations. The endogenous grid method was not used, as noted in the problem statement—the non-convexity introduced by fixed costs makes this approach ill-suited here. The discrete grid approximation means that policy functions are step functions rather than smooth; this is particularly relevant for the inaction region, where the true boundary lies between grid points.

9 Conclusion

We solved and calibrated a heterogeneous firm model with convex adjustment costs, fixed costs, and partial irreversibility. The model successfully replicates the qualitative features of plant-level investment documented by Cooper and Haltiwanger (2006): lumpy investment, inaction, and asymmetry between positive and negative adjustment. The SMM estimation points to moderate convex costs ($\hat{\gamma} \approx 1$) and near-zero fixed costs, though the moment fit reveals a tension between matching the low negative investment fraction and other moments simultaneously—a limitation of the three-parameter specification.

The policy analysis shows that a 10% investment subsidy generates large long-run aggregate effects: capital rises 27.7%, output rises 18.9%, and the subsidy cost is only 2.41% of output. The dominant mechanism is the near-complete elimination of inaction: by lowering the effective purchase price of capital, the subsidy induces previously inactive firms to invest. This extensive margin response is large relative to the intensive margin

effect on investment rates, highlighting how adjustment frictions shape the aggregate response to investment policy.